

A Comparative Analysis of 6T, 8T, and 10T SRAM Cells: Performance Evaluation and Robustness Assessment Proving 10T Superiority

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ABSTRACT

Memory architectures are required to meet the conflicting needs of high density, radiation hardness, and ultra-low power operation with the ever-decreasing miniaturization of satellite systems. The architectural evolution of SRAM technology from conventional 6T structures to innovative 10T designs is investigated in this systematic review with an emphasis on evaluating their use for space applications. Although area-efficient, conventional 6T SRAM cells are inherently constrained in space applications due to their compromised read stability at scaled voltages and sensitivity to single-event upsets. Read-decoupling methods, which added layout complexity but enhanced access stability, were developed with the transition to 8T architectures. By applying innovative circuit-level methods that address multiple space-grade constraints simultaneously, modern 10T SRAM architectures represent a significant leap. These include adaptive power gating techniques for leakage reduction, inherent charge-sharing reduction for radiation hardening, and complete read/write path isolation for better noise immunity. The review explores how these architectural innovations maintain compatibility with traditional CMOS processes while allowing for reliable operation in hostile environments. The research demonstrates how 10T SRAMs really break the reliability-power-area tradeoffs

that constrained earlier generations, building upon extensive research work by space agencies and academic institutions. Specific emphasis is given to design techniques that provide intrinsic radiation hardening without any additional shielding mass, which is an important advantage for small satellites. A review of possible future development directions for space-optimised memory is provided at the conclusion of the discussion. These are novel non-volatile SRAM hybrids and 3D integration methods that have the capability to transform onboard data storage in future space systems. **Keywords-** 6T SRAM, 8T SRAM, 10T SRAM, CMOS, low power, read/write stability, space applications, data stability, virtual grounding, leakage reduction.

I. INTRODUCTION

In today's world, satellite communication finds use in every nook and corner. It is employed in the army, broadcasting, disaster tracking, and so on.

Lightweight satellites are being made due to advancements in technology to reduce manufacturing and maintenance costs. Lightweight satellites require high-density memory cells due to their small size. Due to their strong packing density and increased logic capability in computer digital data processing and the satellite control system, SRAM cells are a

good candidate for this application.[1] "Static Random Access Memory," or simply SRAM, is an important component of memory units. SRAMs are made to be fast and low in power consumption.[1] With deep submicron microelectronics, power reduction and minimisation trends have influenced memory system design paradigms considerably. Static Random Access Memory (SRAM) is one of the prominent building blocks employed in modern day digital systems due to its ability to offer high-speed access, easy integration, and pin compatibility with mainstream CMOS processes. SRAM is commonly used in caches, buffer memories, and register files in all computer systems. SRAM has to function for a long time in a high-dose radiation environment during space missions, and radiation effects are bound to have an effect. A type of error that degrades a piece of information or signal is referred to as a soft error.[1] However, with aggressive scaling of the CMOS technology to nanometre regime, conventional SRAM designs, i.e., the 6-transistor (6T) cell, suffer from low static noise margins, high leakage currents, and reduced stability of operation at low supply voltages. These issues include high leakage currents, low static noise margins, and operation instability at low supply voltages. These issues are exacerbated in space by radiation exposure and very high temperatures.

It has long been known that 6T SRAM has its limitations in space applications. Single-event upsets (SEUs), degraded read stability at reduced supply voltages, and high leakage currents are critical hazards to mission-critical operations. New trade-offs in writeability and area overhead were introduced by 8T SRAM schemes, but read stability was also improved by using decoupled access mechanisms. These problems have inspired the development of new 10T SRAM structures using state-of-the-art circuitry for improved radiation hardness, power efficiency,

and operation reliability simultaneously without incurring prohibitively expensive area penalties.

The development of SRAM technology from 6T to 10T designs is discussed systematically in this review paper with emphasis on their applicability to space-grade implementation.

This review attempts to demystify the technological innovation of space-optimized memory systems through a holistic overview of SRAM innovation. It also envisions new technologies that can shape the next generation of onboard memory in space missions, including 3d-stacked and non-volatile SRAM hybrids. Through this review, we highlight the manner in which 10T SRAM structures represent a breakthrough solution to the growing demands of current space missions and not a marginal innovation.

II. LITERATURE REVIEW SUMMARY

[1]In this paper, 180nm 6T, 8T, and 10T CMOS SRAM cells are compared for Internet of Things applications. SRAM cells are designed and simulated in the paper using the Spectre simulator and Cadence Virtuoso tool. The paper is comparing power, noise, and latency of different SRAM designs. The 10T SRAM cell has been seen to provide enhanced cell stability and ultra-low power dissipation over 6T and 8T architectures. The designed SRAM designs are optimized for use in embedded systems, e.g., PCs, smartphones, and portable memory, which are required to operate at high speeds yet consume minimal power.

[2]The article presents a very stable 10T SRAM cell structure with read stability almost comparable to hold stability. The novel 10T cell, in contrast to the traditional 6T cell, improves the read static noise margin (RSNM) by 3.61x at 0.4V. It is also less power-consumer to read (0.58x) and write (0.70x) than the 6T cell. The

write static noise margin (WSNM) decreases by 0.92x.

[3]The design and analysis of a 10T SRAM cell for Internet of Things applications are described in this study. The key points of the paper are as follows:

- Compared to the conventional 6T, TA8T, D2P11T, and 11T SRAM cells, the 10T SRAM cell dissipates 1.33x/1.21x/1.27x/1.43x less read power at 1V supply voltage.
- The write power of the 10T SRAM cell is 1.86x/2.1x/1.79x/2.07x lower than that of the reference cells at 1V supply.
- The 10T SRAM cell's read power varies very little under process fluctuations.

[4]In this paper, a 10T (LP10T) SRAM cell with high write-ability and read stability is proposed. The proposed LP10T SRAM cell possesses a cross-coupled Schmitt-trigger as well as regular inverters and a dedicated read channel to prevent read-disturbance. It performs pseudo differential write operations with write-assist technology via a write bitline and control signal. The proposed design shows significant improvements over the best SRAM cells in RSNM, WSNM, WM, dynamic/leakage power, and read/write delay with marginal area overhead increase.

[6]The paper presents a new RHBD (Radiation-Hardened-By-Design) 10T SRAM cell that is power-efficient as well as highly SEU-resistant. The main points are that the presented SRAM cell is SEU-resistant, i.e., it recovers from radiation-caused bit flips from 0 to 1 and 1 to 0. Besides being able to achieve this SEU tolerance, it is less power-hungry compared to previous RHBD SRAM designs. A 45nm CMOS technology is used in the design and simulation of the SRAM cell. The presented RHBD 10T SRAM cell can find

application in aerospace and other applications where low power dissipation and good radiation tolerance are essential.

[7]Minimization of power with analysis and design of SRAM cells under low-power schemes are explained in the paper. The paper explains enhancing power efficiency under various temperature conditions using 8T and 10T SRAM cell designs. The paper explains SRAM cell power minimization techniques like write driver logic, charge recycling logic, and precharge scaling. Various temperature conditions in low-power usage can be improved by the 10T SRAM cell with precharge scaling since simulation results reveal that it outperforms the 8T SRAM cell, particularly at high temperatures.

[8]We present in this paper a new 10T SRAM cell configuration called STS-10T with improved soft error immunity for space and ground radiation. The new configuration provides improved read and write stability, access time, critical charge, and overall performance metrics compared to the existing radiation-hardened 10T SRAM cell configurations. The STS-10T configuration has the capability to recover from single-event and multi-node upsets and therefore is an appropriate choice for memory devices for applications in extreme radiation environments.

[9]The paper proposes a low power dissipation and 10T SRAM cell with high static noise margin (SNM) in comparison with the traditional 6T SRAM cell. The most significant aspects of the proposed methodology are: 1) Use of a single bit line for read and write operations, which conserves dynamic power dissipation; 2) Use of sleep transistors to conserve static power; and 3) Use of a dynamic feedback loop to improve the SNM. The proposed 10T SRAM cell is observed to dissipate 83.27% less power for write '0' operation and 85.9% less power for write '1'

operation compared to the 6T SRAM, and with 4.9 times improved SNM.

[11]This paper introduces a quad-node ten transistor (10T) SRAM cell, referred to as the Quatro-10T cell, that provides differential read operation for strong sensing against soft errors. The new cell provides improved noise margin in the sub-0.45V regime and 26% reduced leakage current compared to the conventional 12T DICE SRAM cell. The Quatro-10T cell, in comparison with a conventional 6T SRAM cell, provides comparable noise margin at half supply voltage, which saves a lot of leakage power. Acute neutron radiation tests of a 32-kb SRAM in 90-nm CMOS technology demonstrate that the Quatro-10T cell provides 98% reduced soft error rate compared to the 6T cell.

[12]This article proposes a radiation-hardened memory-by-design 10T (RHMD10T) SRAM cell for application in space technology. The RHMD10T cell experiences improved read delay, read stability, hold power, and soft-error tolerance compared to other existing radiation-hardened SRAM cells. The article also proposes a new performance metric called charge-SNM to power-area-delay (QSPAD) to compare the overall performance of the SRAM cells. The outcome indicates that the proposed RHMD10T cell has the highest value of QSPAD, demonstrating its improved performance compared to the other cells under investigation.

[13]The work suggests a 10T SRAM cell design with less read power dissipation yet with excellent performance and stability. The suggested topology provides differential read and single-ended write operation with improved read margin and write margin over a traditional 6T SRAM cell. In addition to that, the read delay and write delay of the suggested topology are greatly improved over the traditional 6T SRAM

cell. The suggested 10T SRAM cell is optimized to possess minimum power delay product during read operation and lower standby and read power over the traditional 6T SRAM.

[15]The paper proposes a novel 10T SRAM cell design with outstanding power and stability improvement over conventional 6T and sub-threshold 10T SRAM cells. Some of the key points are: The new cell saves the total power by 98% and static power by 55% compared to sub-threshold 10T SRAM. It also improves stability parameters like SINM, SVN_M, WTI and WTV by 91%, 47%, 85% and 53% respectively compared to sub-threshold 10T SRAM. The cell uses a diode-connected NMOS transistor to lower the supply voltage and an additional signal-driven NMOS transistor to lower static power. Extra PMOS transistors are used to improve the read stability at low supply voltages. [16]A very stable and robust 10T SRAM cell is proposed in this paper for low-power portable devices. The key features of the proposed design are: 1) Combining a normal inverter with a Schmitt-trigger inverter in the cell core to enhance hold stability, 2) Using separate read and write paths to improve read and write stability, 3) Implementing a feedback-cutting scheme to eliminate the writing '1' issue, and 4) Reducing dynamic and leakage power consumption using the single-ended structure and stacked transistors in the cell core. Simulation results show that the proposed 10T SRAM cell achieves the highest hold, read, and write stability values among other state-of-the-art designs, with improved power efficiency.

[17]This work presents a comparative analysis of 6T and 8T SRAM cells for 90nm CMOS technology. It compares the power dissipation, delay, static noise margin, and area of the SRAM cells. The results are: 6T SRAM dissipates 254μW of power, has a delay of 4.2ns, and an SNM of

0.4V, and has an area of $0.1458\mu\text{m}^2$. 8T SRAM dissipates $101\mu\text{W}$ of power, has a delay of 5ns, and an SNM of 0.4V, and has an area of $0.1782\mu\text{m}^2$. The work concludes that the 8T SRAM design is more power efficient than the 6T SRAM, but at the cost of a small increase in area and delay.

[19]This paper describes the design and performance analysis of a 1 KB (32×32) SRAM memory array based on a 6T SRAM cell in 22 nm CMOS technology. The new SRAM array had the lowest leakage current of 18.65 pA and the average delay was 19 ns. The read and write power dissipation of the new SRAM array were $48.22\mu\text{W}$ and $385\mu\text{W}$ respectively, better than current 8T SRAM and 7T SRAM designs at power dissipation. The functionality of the 1 KB SRAM array was verified using the Cadence Virtuoso tool in 22 nm technology.

[21]The paper proposes a new 10T SRAM cell named PP10T with the capability of recovery from soft errors. PP10T is single-event upset (SEU)-robust with full-type resistance and partial resistance to multi-node upsets with charge-sharing effects. PP10T is of low area, low power consumption, and read-disturbance-free. Simulation results indicate that PP10T outperforms other radiation-hardened SRAM cells in write/read access time, static noise margin, and hold power consumption. PP10T can also recover from SEUs on single nodes and simultaneous SEUs on multiple nodes.

[24]In this paper, an energy-efficient single-ended 10T (SE10T) near-threshold SRAM is proposed. The proposed SE10T improves read stability and writability with read-assist technique and power-gating method, and conserves power/energy consumption with single-ended read/write and transistor stacking. Simulation results in 32-nm CMOS indicate that

the proposed SE10T achieves significantly improved stability, power, and energy metrics over conventional 6T SRAM, but at the cost of increased area and delay.

[25]We introduce here a highly stable low-energy 10T (SLE10T) SRAM cell design enabling near-threshold operation. One-bitline architecture reduces energy and area; isolated read path and read-assist scheme avoid read-disturbance; tri-state inverter enhances writability; and transistor stacking reduces leakage power. These are the key features of the proposed design. The proposed SLE10T outperforms current SRAM designs in 7-nm FinFET technology at 0.5 V in terms of read stability, writability, read/write energy, and leakage power.

[26]A 10T SRAM cell that is low-power, fast, and robust is designed and applied in this paper. Compared with traditional and other 10T SRAM cells, the designed 10T SRAM cell is likely to improve read performance, reduce area and power, and significantly lower data-dependent read-port leakage. The authors compare the proposed 10T SRAM cell with other architectures through simulation of a 128-kb SRAM and observe improvements in read access efficiency, energy saving, and area saving.

[31]This paper provides a comprehensive analysis of 6T, 8T, and 10T SRAM cell configurations implemented in both CMOS and FinFET technologies. The key findings are:

- FinFET-based SRAM performs better than CMOS SRAM in terms of static noise margin (SNM), propagation delay, and power consumption.
- Compared to the 6T SRAM, the 8T SRAM design exhibits notable advantages in power dissipation and SNM because to

the segregated read and write circuitry.

- The 8T SRAM configuration shows significant improvements in power dissipation and SNM compared to the 6T SRAM, due to the isolated read and write circuitry.
- The 10T SRAM has better read stability and lower power dissipation, but covers a larger cell area compared to the 6T and 8T configurations.
- The selection of the optimal SRAM cell design depends on the specific application requirements, balancing trade-offs between area, efficiency, performance, and radiation hardness.

III. COMPARISON OF 6T, 8T AND 10T SRAM PERFORMANCE

A. Read/ Write Stability Analysis

The common path between the internal storage node and the bit lines causes the 6T SRAM to be plagued with compromised read stability during read access. A probable voltage divider effect resulting from this reduces the Static Noise Margin (SNM) and raises the tendency towards disturb to read disturb errors.[1][11] The internal storage node of the 6T SRAM cell could be vulnerable to bit flips due to read access due to the read-disturb caused by common read/write paths. Static noise margin (SNM) is constrained by this design, especially with low supply voltages and aggressive

scaling.[15][16].

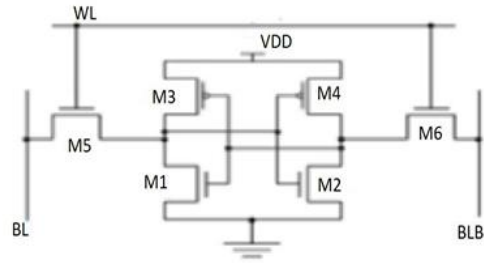


Fig 1. 6T SRAM Cell [31]

By separating the read and write paths—typically through the use of an additional read buffer and word lines—the 8T SRAM improves read stability. Compared to the 6T cell, such read isolation achieves a significant improvement in SNM. Write margins are affected by 8T cells' constant vulnerability to low voltage or high temperature.[7][9] By separating the read and write paths, the 8T SRAM eliminates the read-disturbance. Compared to 6T, this design significantly improves the Read SNM (RSNM). However, with certain PVT variations, the trade-off is negligible write disturbance and area penalty. [17] By including two additional transistors that isolate the read operation from the storage nodes, the 8T SRAM cell alleviates these problems.[31]

It is the reason for an observable improvement in read stability and SNM, from 301.7 mV (6T CMOS) to 376 mV (8T CMOS) and from 320.6 mV (6T FinFET) to 250 mV (8T FinFET).

Improved read and write stability is shown by 10T SRAM, especially by complex designs like the Quatro-10T and feedback-regulated ones. In order to isolate sensitive nodes during operation, certain designs make use of sleep transistors, dynamic feedback, and dedicated read and write ports. These configurations offer robust data retention in read and hold modes and $4.9\times$ more SNM than in 6T.[8][9][11] Significant gains in read and write stability are demonstrated by 10T SRAM designs:

- Internal node disturbance during operations is eliminated by using dynamic feedback techniques and distinct read/write paths.
- More recent designs combine feedback-cutting transistors and Schmitt-trigger-based inverters to improve RSNM and WSNM at the same time.[16] In a situation where 6T fails without increasing the cell ratio, one design achieved $4.9\times$ higher SNM than 6T and dependable operation even at 0.3 V supply.[9][15]

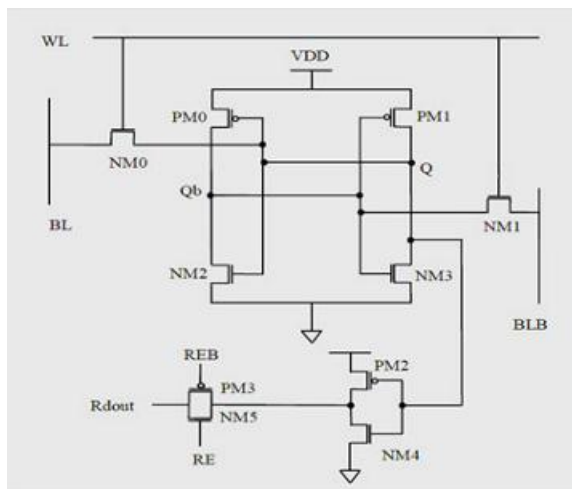


Fig. 2. 10T SRAM Cell [1]

By employing sophisticated design strategies like differential read ports, pseudo-random feedback paths, and VGND switching, the 10T SRAM cell goes one step further. These characteristics greatly improve robustness and noise tolerance. FinFET-based 10T SRAM has a lower SNM (124.9 mV), but it is more resistant to radiation-induced bit flips and process variation.[31] By incorporating NMOS gating or diode-connected feedback isolation, advanced 10T variants further improve write "1" stability, which is frequently the weakest point in single-ended architectures.[16]

B. Power Consumption and Leakage Comparison

While 6T SRAM cells consume moderate dynamic power, they leak in idle mode, which is especially undesirable in scaled technologies. Subthreshold operations exacerbate leakage.[7] Energy efficiency in SRAM design becomes critical, especially for mission-critical and battery-powered systems. Due to its constant path of current in hold and switching modes, the 6T SRAM consumes peak power both in CMOS (94.1 nW) and FinFET (46.4 nW) implementations.[31]

Although 8T SRAM has overhead since it has additional transistors, it supports methods like charge recycling and write driver logic that weaken dynamic power to a certain degree in many situations. Static power and area also increase to a very small degree.[7] Because it has independent read paths and lower bitline activity, the 8T SRAM uses much less power: 35.6 pW in CMOS and only 1.73 nW in FinFET.[31]

10T SRAM power performance is great. Static power and dynamic power are greatly minimized by single-bit-line styles and strategies like charge recycling and precharge scaling. Of note, some 10T cells have higher thermal stability (power is stable at temperatures above 50°C) and an estimated write power reduction of 83–86% compared to 6T.[7][9]

Despite being error-tolerant, differential 10T versions, like the Quatro-10T, also experience 26% less leakage than hardened 12T cells.[11] The FinFET 10T SRAM consumes the maximum overall power (11.13 μ W), quite possibly due to more complicated control circuitry. Nevertheless, the architecture is very good at minimizing leakage current and enhancing energy

efficiency in idle and read modes, especially under PVT stress.

C. [31] Delay and Speed Metrics

Owing to its smaller size and fewer transistors, the 6T SRAM has the shortest delay behavior. Stability and robustness are traded off with the shortest write and read delays.[1] With the write delay of 286.8 ps in CMOS and 242.4 ps in FinFET technology, the 6T SRAM is the fastest. [31]

Since 8T SRAM also has larger bitline capacitances and separated read/write paths, it contributes a moderate delay. In most applications, the delay is still fine despite being larger, particularly in the case of the write operation.[7] The 8T SRAM is relatively fast—424.5 ps for CMOS and 237 ps for FinFET—despite being marginally slower because of the extra transistors and path segregation. [31]

10T SRAM takes more delay although it is more power-efficient and has greater SNM. For example, due to the extra logic control and feedback paths, read delay is greater and writes are about 92 ps (compared to 31 ps for 6T). Improved noise margins and hardness, especially in radiation-hard or ultra-low power environments, often make such delays tolerable.[9][8] Surprisingly, the 10T SRAM features a decent write delay of 140.9 ps in FinFET and a significantly faster write delay of 89.0 ps in CMOS. This is attributed to better control over switching dynamics and efficient signal routing. 10T is ideal for applications requiring reliability over pure access speed since, although complex, it achieves a balance between speed and increased robustness.[31]

V. Recent Advancements and Opportunities

A. Integration of 3D SRAM
Integrating 3D SRAM architectures is a promising approach to greatly reduce interconnect delays while boosting memory density and bandwidth. Using monolithic 3D integration or through-silicon vias (TSVs) to vertically stack multiple SRAM layers, designers can create compact, high-performance cache and buffer solutions for space-constrained applications such as small satellites and edge AI devices. Additionally, the shorter vertical interconnects reduce latency and power dissipation in comparison to planar designs. Future 3D SRAM designs incorporate radiation-hardened cells, such as 10T variants. SEU's resilience is enhanced by using physical isolation in combination with intrinsic hardening. Research into optimising the yield and addressing the thermal management challenges associated with multilayer stacking is necessary for a stable space-grade deployment.

B. Hybrid Non-Volatile SRAM

New hybrid memory cells deliver instant-on memories with ultra-low standby power through the integration of SRAM and non-volatile memory (NVM) devices such as resistive RAM (ReRAM), magnetoresistive RAM (MRAM), or phase-change memory (PCM). Hybrid SRAM cells meet the bare minimum of power-efficient data storage in satellite payloads with unstable power supplies by storing data without electricity. NVM layers or NVM devices built into standard 10T SRAM cells are introduced into typical hybrid implementations to enable fast write-back and recovery after power-down. Long-duration space missions, where energy conservation and data storage against radiation stress are of maximum importance, are most impacted by this development.

C. AI-Aided Memory Design and Optimisation

SRAM design optimisation and automation flows are evolving with developments in artificial intelligence and machine learning (AI/ML) of the recent years. Attempting to balance power, speed, and ruggedness more optimally than manual optimisation, AI-powered tools explore vast design spaces and optimise transistor size, layout, floorplanning, and margining. Attempting to deliver SRAM cells for targeted application areas, e.g., radiation-hardened space technology or ultralow-power Internet of Things devices, surrogate model- and reinforcement learning-based algorithms accelerate exploration of transistor-level trade-offs. Design cycles are minimised, yield prediction is enhanced, and adaptive memory structures that can be customised for evolving system requirements are made possible by this paradigm shift.

D. New Transistor Technologies: GAA and FinFET

SRAM cells are moving toward more advanced transistor structures, including gate-all-around (GAA) nanosheet transistors and FinFETs, that enable new possibilities for enhanced radiation tolerance, reduced leakage, and scaling noise margins. FinFET SRAM designs enable lower supply voltages with less sacrifice in robustness through shorter-channel effects and enhanced drive strength. GAA transistors enable high-density, low-consumption SRAM arrays for hostile environments through additional leakage and variability minimization through full channel wrapping. 10T SRAM cells based on these advances include improved stability and energy efficiency, which makes them suitable for edge computing and future aerospace applications.

IV. CONCLUSION

This systematic review was targeting the use of 6T, 8T, and 10T SRAM cells in space-grade applications through systematic evaluation of their performance, stability, power efficiency, and reliability. The study emphasized how the conventional 6T SRAM is less favorable in aggressive environments like space due to its inherent limitations, such as low read stability, susceptibility to single-event upsets (SEUs), and high leakage currents. Although 8T SRAM structures had compromises in writeability and area overhead, they improved read stability through decoupled access mechanisms.

By addressing the underlying issues of radiation hardness, power efficiency, and operation reliability simultaneously, 10T SRAM architectures, nevertheless, proved to be an excellent solution. By implementing advanced methods like adaptive power gating, dynamic feedback, and differential read ports, cutting-edge 10T designs exhibited significant improvements in SNM, lower power consumption (up to 83–86% lower write power than 6T), and better immunity against SEUs. For these reasons, 10T SRAM cells are most appropriate for space applications with low power consumption and reliability as the top priorities.

The review also pointed out the ways in which new technologies like 3D integration and hybrid non-volatile SRAMs would transform onboard memory systems in subsequent space missions. 10T SRAM architectures are a new memory technology for hostile environments by realizing an ideal balance among the area, speed, and reliability tradeoff.

In short, 10T SRAM is the optimal selection for next-generation space systems due to its record-breaking radiation resistance, power efficiency, and reliability. The implementation will be critical in meeting the growing demands of modern space exploration and satellite missions.

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